QRYVANTA Quiz Bank

63 original questions | Answers and explanations | Difficulty: Easy / Medium / Hard

This document is generated from the same QUESTION_BANK as the in-app offline quiz. The document preserves canonical question/option order for study; the app randomizes question and option order in each attempt. These are original practice questions, not claimed past-paper questions.

How to study

Try each question before checking the answer. Use the explanation to revise the relevant idea or rule ID. Performance-rule questions describe the current QRYVANTA source implementation and the metadata it relies on.

1. [Q001] Which clause filters individual rows before grouping?

Topic: SQL basics | Difficulty: Easy

A. WHERE

B. HAVING

C. ORDER BY

D. SELECT

Correct answer: WHERE

Explanation: WHERE filters source rows; HAVING filters groups after grouping.

2. [Q002] Which clause filters aggregate groups?

Topic: SQL basics | Difficulty: Easy

A. HAVING

B. WHERE

C. FROM

D. VALUES

Correct answer: HAVING

Explanation: HAVING is applied to grouped results, usually with aggregate conditions.

3. [Q003] What is the correct predicate for rows where end_date is missing?

Topic: NULL semantics | Difficulty: Easy

A. end_date IS NULL

B. end_date = NULL

C. end_date == NULL

D. end_date <> NULL

Correct answer: end_date IS NULL

Explanation: Use IS NULL; equality/inequality comparisons with NULL evaluate to UNKNOWN.

4. [Q004] Which statement changes existing rows?

Topic: SQL basics | Difficulty: Easy

A. UPDATE

B. INSERT

C. CREATE

D. SELECT

Correct answer: UPDATE

Explanation: UPDATE modifies values in existing rows.

5. [Q005] What is the main risk of UPDATE table SET flag = 1 with no WHERE?

Topic: Safe writes | Difficulty: Easy

A. It can update every row in the table

B. It creates a new table

C. It always updates exactly one row

D. It cannot execute in SQL

Correct answer: It can update every row in the table

Explanation: Without a predicate, the statement applies to all rows unless other database mechanisms restrict it.

6. [Q006] Which clause determines the order returned to the client?

Topic: SQL basics | Difficulty: Easy

A. ORDER BY

B. GROUP BY

C. WHERE

D. JOIN

Correct answer: ORDER BY

Explanation: ORDER BY defines result ordering; without it, row order is not guaranteed.

7. [Q007] What is a common benefit of an index?

Topic: Index basics | Difficulty: Easy

A. It can help locate qualifying rows without scanning every row

B. It guarantees every query is faster

C. It removes the need for statistics

D. It prevents all duplicate values

Correct answer: It can help locate qualifying rows without scanning every row

Explanation: Indexes offer access paths but can have storage/write costs and are not always selected.

8. [Q008] Which join returns only matching pairs under its join predicate?

Topic: JOINs | Difficulty: Easy

A. INNER JOIN

B. FULL OUTER JOIN

C. CROSS JOIN

D. LEFT JOIN

Correct answer: INNER JOIN

Explanation: INNER JOIN returns row pairs satisfying the join condition.

9. [Q009] What does AST stand for?

Topic: AST | Difficulty: Easy

A. Abstract syntax tree

B. Automatic SQL timer

C. Application storage table

D. Access statistics trace

Correct answer: Abstract syntax tree

Explanation: An AST represents parsed statement structure as nodes.

10. [Q010] Which artifact shows the database's chosen operators and access paths?

Topic: Query plans | Difficulty: Easy

A. Execution plan

B. CSV export

C. Glossary

D. Account profile

Correct answer: Execution plan

Explanation: An execution plan describes the physical strategy selected by the optimizer.

11. [Q011] Which statement best describes static linting?

Topic: Linting | Difficulty: Easy

A. It checks query structure for known patterns without requiring elapsed-time measurement

B. It proves a query will be fastest

C. It always executes a query twice

D. It replaces database authentication

Correct answer: It checks query structure for known patterns without requiring elapsed-time measurement

Explanation: Static rules can flag patterns from parsed SQL; they do not prove actual runtime.

12. [Q012] What does SQL001 flag in QRYVANTA?

Topic: SQL001 | Difficulty: Easy

A. A SELECT projection using a bare *

B. Every COUNT(*) aggregate

C. Any WHERE clause

D. Every index scan

Correct answer: A SELECT projection using a bare *

Explanation: SQL001 checks for an exp.Star in SELECT expressions, distinct from COUNT(*).

13. [Q013] Which predicate should replace status = NULL?

Topic: SQL015 | Difficulty: Easy

A. status IS NULL

B. status == NULL

C. status <> NULL

D. status LIKE NULL

Correct answer: status IS NULL

Explanation: SQL015 flags equality/inequality AST nodes compared with NULL.

14. [Q014] Which predicate is generally more likely to use a regular index on created_at?

Topic: SARGability | Difficulty: Easy

A. created_at >= :start AND created_at < :next_start

B. YEAR(created_at) = 2026

C. TO_CHAR(created_at, 'YYYY') = '2026'

D. TRUNC(created_at) = :day

Correct answer: created_at >= :start AND created_at < :next_start

Explanation: A half-open range leaves the indexed column bare; confirm equivalent time semantics for the database.

15. [Q015] Why is ORDER BY important with LIMIT/FETCH?

Topic: Pagination | Difficulty: Easy

A. It makes the selected subset deterministic

B. It always avoids all sorting

C. It automatically creates an index

D. It grants table permissions

Correct answer: It makes the selected subset deterministic

Explanation: Without ordering the database is free to return any qualifying subset.

16. [Q016] What is the safest way to handle an untrusted value in an application query?

Topic: Database safety | Difficulty: Easy

A. Bind it as a parameter

B. Concatenate it into SQL text

C. Remove all spaces manually

D. Place it inside a comment

Correct answer: Bind it as a parameter

Explanation: Parameterized statements separate values from SQL syntax and are a key injection defense.

17. [Q017] What risk does SQL003 identify?

Topic: SQL003 | Difficulty: Medium

A. DELETE with no WHERE node

B. DELETE with a valid filter

C. SELECT with ORDER BY

D. A table with no index

Correct answer: DELETE with no WHERE node

Explanation: The source rule checks for a DELETE node without a WHERE clause.

18. [Q018] Which LIKE pattern triggers SQL006's leading-wildcard check?

Topic: SQL006 | Difficulty: Medium

A. email LIKE '%@example.com'

B. email LIKE 'sam%'

C. email LIKE 'abc_9'

D. email LIKE 'x'

Correct answer: email LIKE '%@example.com'

Explanation: The rule flags a literal pattern starting with % or _.

19. [Q019] Why can NOT IN (subquery) behave unexpectedly if the subquery returns NULL?

Topic: SQL007 | Difficulty: Medium

A. Three-valued logic can make the predicate UNKNOWN

B. NULL becomes zero

C. NOT IN always turns into an INNER JOIN

D. The query skips parsing

Correct answer: Three-valued logic can make the predicate UNKNOWN

Explanation: A NULL among NOT IN values can prevent rows from evaluating TRUE.

20. [Q020] When does SQL008 flag a literal OFFSET?

Topic: SQL008 | Difficulty: Medium

A. At 1000 or more

B. At any positive number

C. Only when OFFSET is zero

D. Only if ORDER BY is present

Correct answer: At 1000 or more

Explanation: The source rule ignores non-literal offsets and values below 1000.

21. [Q021] What does SQL009 warn about?

Topic: SQL009 | Difficulty: Medium

A. A limited/fetched result without ORDER BY

B. Every SELECT with WHERE

C. Every join with ON

D. A DELETE with RETURNING

Correct answer: A limited/fetched result without ORDER BY

Explanation: The rule warns that pagination without a specified order is nondeterministic.

22. [Q022] What is a likely consequence of an accidental Cartesian product?

Topic: SQL010 | Difficulty: Medium

A. Rows from the two sources multiply

B. The tables are automatically indexed

C. NULLs are removed

D. The query becomes read-only

Correct answer: Rows from the two sources multiply

Explanation: Without a relationship predicate, combinations of rows can multiply output cardinality.

23. [Q023] Why should INSERT specify a target column list?

Topic: SQL011 | Difficulty: Medium

A. It avoids relying on physical/defined column order

B. It disables constraints

C. It guarantees a faster plan

D. It commits the transaction

Correct answer: It avoids relying on physical/defined column order

Explanation: An explicit column list makes value-to-column mapping robust to schema changes.

24. [Q024] What pattern does SQL012 inspect?

Topic: SQL012 | Difficulty: Medium

A. OR branches referring to different column sets

B. AND between identical values only

C. GROUP BY without aliases

D. Every IS NULL predicate

Correct answer: OR branches referring to different column sets

Explanation: The code compares the column-name sets on the left and right of OR nodes in WHERE.

25. [Q025] Which Oracle expression is checked as a random-ordering signal?

Topic: SQL013 | Difficulty: Medium

A. DBMS_RANDOM in ORDER BY

B. CURRENT_DATE in WHERE

C. COUNT(*) in SELECT

D. ROWNUM in WHERE

Correct answer: DBMS_RANDOM in ORDER BY

Explanation: The rule has dialect-specific checks for random functions and extra Oracle DBMS_RANDOM handling.

26. [Q026] Why can a correlated scalar subquery in SELECT be costly?

Topic: SQL014 | Difficulty: Medium

A. It may do additional work for each outer result row

B. It necessarily deletes data

C. It forbids all indexes

D. It automatically sorts by primary key

Correct answer: It may do additional work for each outer result row

Explanation: The source rule flags a projection expression containing a Subquery node.

27. [Q027] For a B-tree index on (region, created_at), which filter best matches its leading key?

Topic: Index basics | Difficulty: Medium

A. region = 'NE' AND created_at >= :d

B. created_at >= :d only

C. LOWER(region) = 'ne' only

D. A filter on an unrelated column only

Correct answer: region = 'NE' AND created_at >= :d

Explanation: The region predicate uses the leading key and then can constrain the second key.

28. [Q028] Why might an index on a two-value status column be unattractive alone on a huge table?

Topic: Selectivity | Difficulty: Medium

A. Each value may match a large fraction of rows

B. Indexes cannot contain strings

C. The database cannot compare equality

D. It prevents joins

Correct answer: Each value may match a large fraction of rows

Explanation: Low cardinality often means low selectivity, although the optimizer's full cost model matters.

29. [Q029] What should you do when estimated rows differ greatly from actual rows?

Topic: Query plans | Difficulty: Medium

A. Investigate statistics, skew, correlations and parameter values

B. Assume the SQL parser is wrong

C. Add every possible index

D. Ignore the difference

Correct answer: Investigate statistics, skew, correlations and parameter values

Explanation: Cardinality errors can change join order and access-path decisions.

30. [Q030] A nested-loop join is often a good fit when...

Topic: JOINs | Difficulty: Medium

A. The outer input is small and the inner lookup is efficient

B. Both sides must be billions of rows

C. The join condition is missing

D. All indexes must be dropped

Correct answer: The outer input is small and the inner lookup is efficient

Explanation: Nested loops can be effective for small outer sets and indexed probes.

31. [Q031] Which is the strongest way to validate a proposed optimization?

Topic: Performance workflow | Difficulty: Medium

A. Compare correctness, plans and measurements on representative data

B. Use the shortest SQL text

C. Assume every warning requires an index

D. Compare only variable names

Correct answer: Compare correctness, plans and measurements on representative data

Explanation: Plans, semantics and measured workload behavior all matter.

32. [Q032] What evidence is combined by P002?

Topic: P002 | Difficulty: Medium

A. A filtered table scan and no leading index on the filtered column

B. Any SELECT * statement

C. A missing semicolon

D. A random ORDER BY function only

Correct answer: A filtered table scan and no leading index on the filtered column

Explanation: The rule checks a bare filter, table scan and index metadata.

33. [Q033] What does P003 report?

Topic: P003 | Difficulty: Medium

A. A sort operation in the execution plan

B. A user login attempt

C. A missing CREATE TABLE

D. A database schema with no views

Correct answer: A sort operation in the execution plan

Explanation: The source rule fires if the normalized plan reports sorts.

34. [Q034] Which condition can be marked non-SARGable by the source context?

Topic: P005 | Difficulty: Medium

A. A function or arithmetic expression wraps a WHERE column

B. A bare equality on an indexed key

C. A SELECT list alias

D. A parameter bound separately

Correct answer: A function or arithmetic expression wraps a WHERE column

Explanation: The context records wrapped predicates for performance checks.

35. [Q035] When can P006 fire?

Topic: P006 | Difficulty: Medium

A. A join uses a scanned table whose join column lacks a detected supporting index

B. Any query contains ORDER BY

C. A table has a primary key

D. The query contains only a literal

Correct answer: A join uses a scanned table whose join column lacks a detected supporting index

Explanation: P006 needs join pairs, a join plan, a scanned table and no detected index on the column.

36. [Q036] What comparison does P008 target?

Topic: P008 | Difficulty: Medium

A. A numeric literal compared with a column typed as a string

B. A string literal compared with a string column

C. A NULL check

D. A BETWEEN with date parameters only

Correct answer: A numeric literal compared with a column typed as a string

Explanation: The rule uses metadata and numeric-literal detection for configured string types.

37. [Q037] What does a left-prefix index finding mean?

Topic: P009 | Difficulty: Medium

A. One index's columns match the start of a longer index

B. The shorter index has more columns

C. Every index is invalid

D. The optimizer used a full-text search

Correct answer: One index's columns match the start of a longer index

Explanation: P009 excludes primary/unique shorter indexes and labels the finding as potential redundancy.

38. [Q038] For index (a,b,c), filtering only on b may be problematic because...

Topic: P010 | Difficulty: Medium

A. The normal B-tree access path usually starts with leading column a

B. B-trees cannot index b

C. b is always NULL

D. Indexes cannot have three columns

Correct answer: The normal B-tree access path usually starts with leading column a

Explanation: A suffix-only predicate does not match the composite index's leading key.

39. [Q039] What combination is needed for P012?

Topic: P012 | Difficulty: Medium

A. GROUP BY column lacks a leading index and the plan contains a sort/grouping operation

B. Any query without GROUP BY

C. A primary-key lookup only

D. A view with no columns

Correct answer: GROUP BY column lacks a leading index and the plan contains a sort/grouping operation

Explanation: The code checks group columns, supporting indexes, and sorting/grouping plan evidence.

40. [Q040] Why can updating an indexed column increase write work?

Topic: P014 | Difficulty: Medium

A. The corresponding index entries must be maintained

B. Indexes do not change on updates

C. It forces a new schema

D. It disables transaction logs

Correct answer: The corresponding index entries must be maintained

Explanation: P014 detects UPDATE assignments whose target columns are indexed.

41. [Q041] What metadata does P015 compare on join columns?

Topic: P015 | Difficulty: Medium

A. Data types and, when both exist, collations

B. Column display names only

C. File paths

D. Theme colors

Correct answer: Data types and, when both exist, collations

Explanation: The code checks data type inequality and non-empty collation mismatch.

42. [Q042] What is the concern behind a join row explosion?

Topic: P017 | Difficulty: Medium

A. The join estimate is much larger than child estimates

B. Every join returns zero rows

C. All joined columns have indexes

D. The query has a comment

Correct answer: The join estimate is much larger than child estimates

Explanation: P017 checks join output estimates against the largest available child estimate and configured thresholds.

43. [Q043] What is materialization in a plan?

Topic: P019 | Difficulty: Medium

A. An intermediate result is produced for later operations

B. The parser deletes the AST

C. The connection is logged out

D. The table becomes immutable

Correct answer: An intermediate result is produced for later operations

Explanation: Temporary/materialized operations can add memory, I/O and CPU overhead.

44. [Q044] At what estimated fraction does P020 flag large index reads?

Topic: P020 | Difficulty: Medium

A. At least 30% of known table rows

B. At least 1%

C. Exactly 100% only

D. Any nonzero row count

Correct answer: At least 30% of known table rows

Explanation: The code tests estimated index rows divided by known table row count >= 0.30.

45. [Q045] What does a low-cardinality index signal?

Topic: P021 | Difficulty: Medium

A. Its leading key has few distinct values on a large table

B. It has too many pages in the UI

C. It cannot be queried

D. It has only NULL values by definition

Correct answer: Its leading key has few distinct values on a large table

Explanation: The rule uses known leading-column cardinality and table size thresholds.

46. [Q046] Why is an AST generally more robust than searching the whole SQL string for `*`?

Topic: AST | Difficulty: Hard

A. It can distinguish a projection Star node from a star inside COUNT(*)

B. It guarantees no parsing errors

C. It measures network latency

D. It replaces dialect selection

Correct answer: It can distinguish a projection Star node from a star inside COUNT(*)

Explanation: AST node types distinguish structural contexts that can look textually similar.

47. [Q047] When replacing OR branches with UNION ALL, what must be checked?

Topic: SQL semantics | Difficulty: Hard

A. Whether duplicates and result semantics remain equivalent

B. Whether the new query uses fewer lines

C. Whether the app theme is dark

D. Whether every branch has a comment

Correct answer: Whether duplicates and result semantics remain equivalent

Explanation: UNION ALL preserves duplicates and may behave differently than an OR predicate in complex cases.

48. [Q048] Why can comparing a string-typed ID column to a numeric literal be risky?

Topic: SARGability | Difficulty: Hard

A. An implicit conversion may alter comparison semantics or index use

B. Strings always sort as numbers

C. It automatically creates an index

D. It prevents NULLs

Correct answer: An implicit conversion may alter comparison semantics or index use

Explanation: Data type mismatch can trigger conversion; use deliberate compatible values and verify behavior.

49. [Q049] Why should P009 not automatically delete the shorter index?

Topic: Indexes | Difficulty: Hard

A. Constraints, workload and write cost may justify keeping it

B. A left-prefix index can never be redundant

C. The shorter index may contain secrets

D. All indexes are required by SQL

Correct answer: Constraints, workload and write cost may justify keeping it

Explanation: P009's own recommendation asks to validate real workload, constraints and write overhead.

50. [Q050] A table scan is likely reasonable when...

Topic: Execution plans | Difficulty: Hard

A. A query reads much of a small table or a low-selectivity result

B. Any WHERE clause exists

C. The query has a typo

D. The table has a name longer than 10 characters

Correct answer: A query reads much of a small table or a low-selectivity result

Explanation: The optimizer may prefer sequential access when much of the table is needed.

51. [Q051] Why can stale statistics harm a plan?

Topic: Statistics | Difficulty: Hard

A. The optimizer's row/cost estimates may misrepresent the data

B. They change SQL keyword spelling

C. They disable all joins

D. They modify application theme settings

Correct answer: The optimizer's row/cost estimates may misrepresent the data

Explanation: Plan selection depends on estimates, distributions and other costs.

52. [Q052] Why does NOT IN with a NULL-containing subquery differ from a simple anti-join intuition?

Topic: NULL semantics | Difficulty: Hard

A. Comparisons involving NULL can evaluate UNKNOWN

B. NULL is treated as the smallest string

C. NOT IN ignores all subquery rows

D. The database converts NULL to false everywhere

Correct answer: Comparisons involving NULL can evaluate UNKNOWN

Explanation: SQL three-valued logic changes whether the predicate evaluates TRUE.

53. [Q053] What is keyset pagination's core idea?

Topic: Pagination | Difficulty: Hard

A. Continue after a stable key value instead of skipping a large offset

B. Read every earlier row and discard it

C. Randomly reorder each page

D. Remove ORDER BY

Correct answer: Continue after a stable key value instead of skipping a large offset

Explanation: Keyset/seek paging avoids work proportional to a very large skipped prefix when indexed correctly.

54. [Q054] Why is adding an index not automatically an optimization?

Topic: Query optimization | Difficulty: Hard

A. It adds storage and write-maintenance costs and may not be selected

B. Indexes cannot speed reads

C. Indexes disable transactions

D. It always changes query results

Correct answer: It adds storage and write-maintenance costs and may not be selected

Explanation: Index decisions require access-path, selectivity, read benefit and write-cost validation.

55. [Q055] Which condition can prevent P011 from suggesting a covering index?

Topic: P011 | Difficulty: Hard

A. The query selects * or the relevant table is known to be below the small-table threshold

B. The query has a filter

C. The index list is empty

D. The app is in day mode

Correct answer: The query selects * or the relevant table is known to be below the small-table threshold

Explanation: The source returns early for SELECT-star and for known small tables.

56. [Q056] Why does P013 consider a LIMIT when selecting a heavy column?

Topic: P013 | Difficulty: Hard

A. Limiting rows can reduce payload I/O and result-transfer cost

B. LIMIT changes the column type

C. A LIMIT creates an index

D. A LIMIT guarantees stable ordering

Correct answer: Limiting rows can reduce payload I/O and result-transfer cost

Explanation: The code flags configured heavy column types selected without a limit.

57. [Q057] An existing index is not used; what is a sound first response?

Topic: P016 | Difficulty: Hard

A. Check selectivity, statistics, conversions and planner costs

B. Always force the index

C. Drop all other indexes

D. Change every column to text

Correct answer: Check selectivity, statistics, conversions and planner costs

Explanation: The rule recommendation specifically asks to inspect these factors before adding another index.

58. [Q058] Which plan evidence triggers P018?

Topic: P018 | Difficulty: Hard

A. A sort estimate at least the large-row threshold with no query limit

B. Any small sort with LIMIT

C. Every index scan

D. A missing database account

Correct answer: A sort estimate at least the large-row threshold with no query limit

Explanation: The rule checks no limit and a large estimated row count for a sort node.

59. [Q059] What extra evidence distinguishes P022 from a generic non-SARGable warning?

Topic: P022 | Difficulty: Hard

A. The wrapped column is indexed and the plan contains a table scan

B. There is an ORDER BY

C. The query has more than one table

D. A SELECT list contains an alias

Correct answer: The wrapped column is indexed and the plan contains a table scan

Explanation: P022 combines predicate wrapping, index metadata and plan scan evidence.

60. [Q060] Which module contains QRYVANTA's static SQL lint rule implementations?

Topic: Architecture | Difficulty: Hard

A. linter/rules.py

B. database_view.py

C. QRYVANTA-Setup.iss

D. assets/qryvanta_logo.png

Correct answer: linter/rules.py

Explanation: The source tree keeps SQL lint checks in linter/rules.py and dispatches them in linter/linter.py.

61. [Q061] Which files chiefly implement the performance rule checks?

Topic: Architecture | Difficulty: Hard

A. analyzer/performance_rules.py and analyzer/analyzer.py

B. QRYVANTA.spec only

C. README.txt only

D. database/seed.py only

Correct answer: analyzer/performance_rules.py and analyzer/analyzer.py

Explanation: Performance rules are defined in performance_rules.py and assembled/orchestrated by analyzer.py.

62. [Q062] Why must `.env` and credentials be excluded from source release ZIPs?

Topic: Security | Difficulty: Hard

A. They can expose private connection settings or secrets

B. They stop Python syntax checks

C. They cannot be read by Python

D. They contain the question bank

Correct answer: They can expose private connection settings or secrets

Explanation: Secrets should stay outside distributable source and use runtime configuration/credential storage.

63. [Q063] Which comparison is most meaningful for a performance rewrite?

Topic: Benchmarking | Difficulty: Hard

A. Same representative workload/parameters, correctness checks, and plan plus elapsed-time comparison

B. Only comparing query character count

C. Only checking if parsing succeeds

D. Comparing unrelated tables

Correct answer: Same representative workload/parameters, correctness checks, and plan plus elapsed-time comparison

Explanation: Performance claims need a repeatable workload and correct results, not syntax aesthetics alone.